You Are My *Quantum* Penguin 

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Playable Prototype Presentation

Brief Game Description

🐧 → explore → 🧊↔🌊 → 🐟 + puzzles → 🪨 → return home → 🐧🪨🐧

- You Are My *Quantum* Penguin is a cozy 2D puzzle-adventure game set in Antarctica.
- The player controls a penguin who goes to explore the world and find something special for their partner.
- Along the way, the penguin catches fish, uses quantum gates, and solves puzzles to reach new areas.
- **Goal:** Explore Antarctica, find the perfect pebble, and return home to your partner.

Implemented Features And Current Gameplay

- Penguin movement and exploration
- Ice and underwater environments
- Fish catching and resource collection
- Quantum gates: X, H, and Z
- Quantum superposition with split-screen visualization
- Quantum phase barrier puzzle
- Partner dialogue system
- Quest objectives and tutorial hints
- Controls overlay
- Original hand-drawn artwork, including characters, environments, and game assets

Current gameplay: Explore → Catch fish → Apply quantum gates → Solve barrier puzzle

Quantum Concepts and Mechanics

The penguin is represented as a qubit with two basis states:

- **|0> - Ice:** The penguin explores the surface.
- **|1> - Water:** The penguin explores underwater.

The player can apply quantum gates using fish as a resource:

- **X gate (1 fish):** Switches between ice and water.
- **H gate (2 fish):** Creates or transforms superposition.
- **Z gate (3 fish):** Changes the relative phase in superposition.

In superposition, the game displays both environments simultaneously using a split-screen view.

How do quantum mechanics affect gameplay?

The prototype includes a quantum phase barrier that requires the state **|->** to open.

The player must:

1. Catch enough fish to power quantum gates.
2. Apply the **H gate** to create superposition.
3. Apply the **Z gate** to change the relative phase.
4. Reach the |-> state and pass through the barrier.

Plans For The Final Version

- More quantum puzzles and challenges
- Additional areas to explore
- The special pebble as a collectible item
- A complete ending: return home and give the pebble to the partner
- Further gameplay balancing and testing

Main priority: Complete the story while expanding the quantum puzzle mechanics.

AI Usage

Tool: ChatGPT

AI was used to assist with:

- Debugging - identifying errors and suggesting possible fixes.
- Code organization - restructuring and organizing code into separate modules.
- Code scaffolding - generating basic code templates and skeletons that were later filled in and customized manually.